



Hot Spots in Our Playground

My name _____

Class _____

Date _____

Where is the hottest place at our school, and how could we cool it down?

Cities can get much hotter than the countryside. Some places at school feel hot and some feel cool. Let's measure them with a micro:bit thermometer!

My kit tick when you have it

- ☐ A micro:bit with a battery pack
- ☐ The thermometer program from Activity 3
- ☐ A sunny spot and a shady spot
- ☐ Hard ground, grass and sand, all in the sun
- ☐ A sheet of black paper
- ☐ A sheet of white paper

Stay safe

- Hold the micro:bit by its edges.
- Keep the battery pack dry.
- Wear a hat and drink water outside.
- Stay where your teacher can see you.
- Do not touch very hot ground for long.

START HERE: BUILD



Get your thermometer ready

1 · DO IT: TICK EACH STEP

- 1 ☐ Plug the battery pack into the micro:bit.
- 2 ☐ Press button B to see the temperature.
- 3 ☐ Put it on your desk. Wait 2 minutes.
- 4 ☐ Press B again and read the number.

2 · LOOK & RECORD

	Temperature in the classroom
Our classroom	°C

WORDS TO USE

temperature degrees warm cool number micro:bit

TEST 1 Sun or shade?

? Is it hotter in the sun or in the shade?

1 · MY GUESS: CIRCLE ONE

- In the sun In the shade Both the same

How sure am I?

2 · DO IT: TICK EACH STEP

- 1 ☐ ☀️ Put the micro:bit on the ground in the sun.
- 2 ☐ 🖐️ Do not hold it. Your hand warms it up.
- 3 ☐ ⌚ Wait 2 minutes, then press B and read it.
- 4 ☐ 🌳 Now do the same in the shade.

3 · LOOK & RECORD

Test	🔌 What does the micro:bit say?
☀️ In the sun	°C
🌳 In the shade	°C

✎ Draw the place: In the sun

✎ Draw the place: In the shade

4 · WHAT HAPPENED? CIRCLE ONE

☒ My guess was right

☐ 🌐 Partly right

☐ 🔄 Something different happened

☐ 🤔 Not sure yet

WORDS TO USE

sun

shade

tree

hot

cool

warmer

cooler

degrees

TEST 2 🟤 Which ground is hottest?

? Which ground gets the hottest in the sun?

1 · MY GUESS: CIRCLE ONE

☐ 🏠 Hard ground

☐ 🌱 Grass

☐ 🏖️ Sand

How sure am I? 🤔 😊 😎

2 · DO IT: TICK EACH STEP

- 1 ☐ 🏠 Lay the micro:bit on hard ground in the sun.
- 2 ☐ ⌚ Wait 2 minutes, then press B and read it.
- 3 ☐ 🌱 Do the same on grass.
- 4 ☐ 🏖️ Do the same on sand.
- 5 ☐ ⚖️ Use the same sunny time for all three.

3 · LOOK & RECORD

Test	🔌 What does the micro:bit say?
🏠 Hard ground	°C
🌱 Grass	°C
🏖️ Sand	°C

 Draw the place: Hard ground

 Draw the place: Grass

 Draw the place: Sand

4 · WHAT HAPPENED? CIRCLE ONE

☒ My guess was right ☐ Partly right ☐ Something different happened ☐ Not sure yet

WORDS TO USE

concrete grass sand hottest coolest sun degrees

TEST 3 Dark or light roof?

? Which colour gets hotter in the sun?

1 · MY GUESS: CIRCLE ONE

☒ Black paper ☐ White paper ☐ Both the same

How sure am I?   

2 · DO IT: TICK EACH STEP

- ☐ Put the micro:bit in the sun under black paper.
- ☐ Wait 5 minutes, then press B and read it.
- ☐ Let it cool, then test under white paper.
- ☐ Use the same spot for both tests.

3 · LOOK & RECORD

Test	 What does the micro:bit say?
☒ Under black paper	°C
☐ Under white paper	°C

 Draw the place: Under black paper

 Draw the place: Under white paper

4 · WHAT HAPPENED? CIRCLE ONE

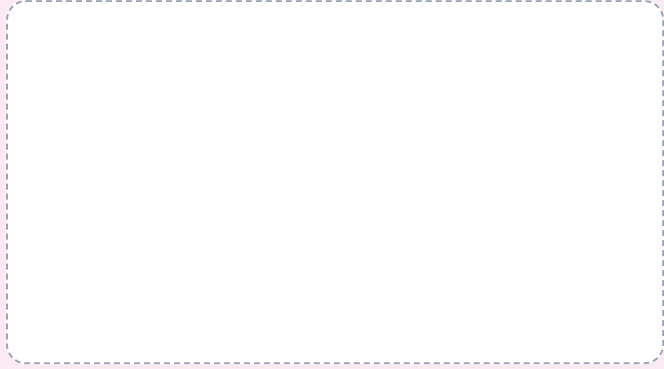
☒ My guess was right ☐ Partly right ☐ Something different happened ☐ Not sure yet

WORDS TO USE

black white dark light roof hotter reflect

Art corner: Cool School map

Draw a map of the playground. Colour hot places red and cool places blue. Draw one thing that would cool a hot place.



Maths corner: How much hotter?

Which number is bigger: sun or shade? Circle it.



New words

Temperature: How hot or cold something is.

Surface: The top or outside of something.

Degrees: The steps we count temperature in.

Reflect: To bounce light back, like a mirror.

Shade: A place where something blocks the sunlight.

What I can do now colour the stars

I can measure temperature with a micro:bit.



I can do a fair test by waiting and not holding the thermometer.



I can suggest a way to make a hot place cooler.

